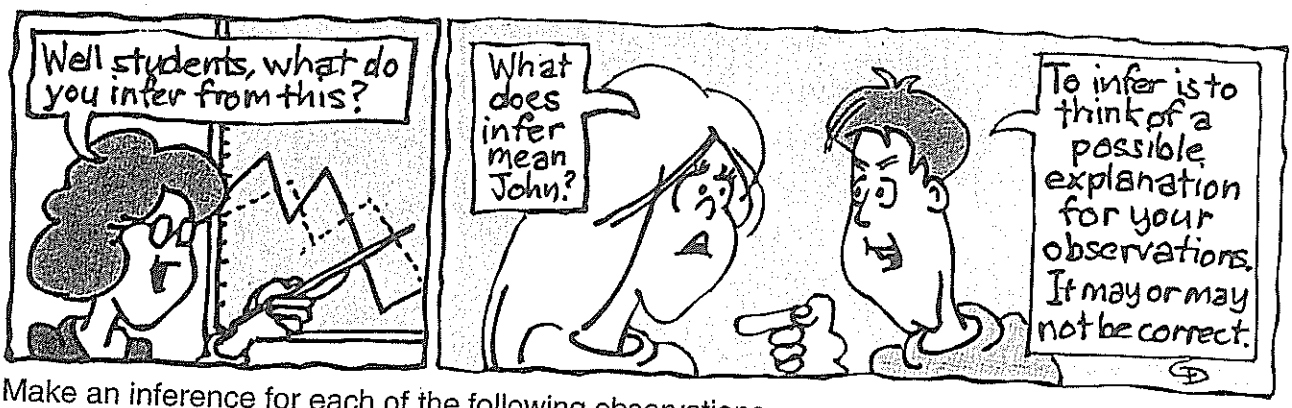
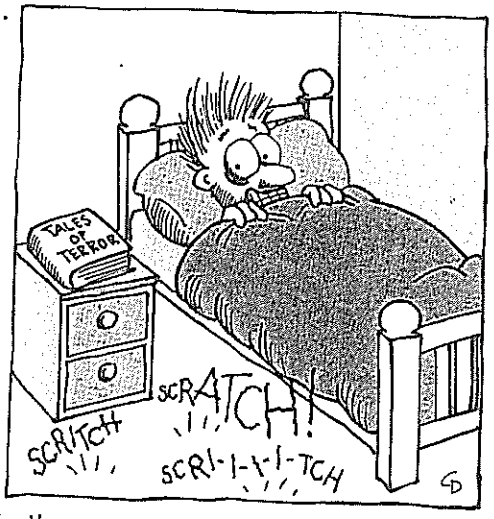
Exercise 3 Observing and inferring

An **inference** is an explanation you can give based on your observations. The inference may or may not be true.



Make an inference for each of the following observations. Compare them with the inferences that other people in your group make.



1 **Observation:** You wake up suddenly in the middle of the night. You hear a faint scratching noise from under the house.

Inference: .....

.....

.....

.....

.....

.....

2 **Observation:** The leaves on the tree have many holes in them.

Inference: .....

.....

3 **Observation:** There is broken glass at the intersection.

Inference: .....

.....

.....

4 **Observation:** When we added acid to the zinc it bubbled vigorously.

Inference: .....

.....

.....

5 **Observation:** The drop of methylated spirits on my hand disappeared.

Inference: .....

.....

.....

6 **Observation:** This rock outcrop on the surface of Mars looks like a human face when lit from the side.

Inference: .....

.....

.....

Exercise 5 Predicting



Complete the table below.

OBSERVATIONS	PREDICTIONS
1 Mickey bought a hot dog for lunch on Monday. 2 Mickey bought a hot dog for lunch on Tuesday. 3 Mickey bought a hot dog for lunch on Wednesday.	....
1 Ants are trekking into the house. 2 The humidity is 100%. 3 The sky is grey and overcast.	....
1 ..... 2 ..... 3	There will be a full moon tomorrow night.

Exercise 6 Handling data

Measurements or observations are called **data**. The best way to record data in a form that is easy to understand is to use a **data table**.

After he reached the age of two, Paul's parents measured his height every year. They recorded these measurements in a data table.

AGE	HEIGHT (cm)
2	88
3	92
4	94
5	94
6	97
7	101

1 How tall was Paul when he was four?

.....

2 How old was he when he was 92 cm tall?

.....

3 How old was he when he reached 1 m in height?

.....

4 *Predict* how tall he will be when he is eight.

.....

2 Select the appropriate caption (the word in **bold type**) from the list and write it below its illustration.

CAPTIONS

- **Observing**—is using your senses to find out as much as you can about an object or an event.
- **Recording**—is writing down your observations or data.
- **Inferring**—is trying to find possible explanations for your observations. (Your inference may or may not be correct.)
- **Predicting**—is using your past observations to say what might happen in the future. (Your prediction may or may not be correct.)
- **Generalising**—is writing a statement that seems true in all cases.



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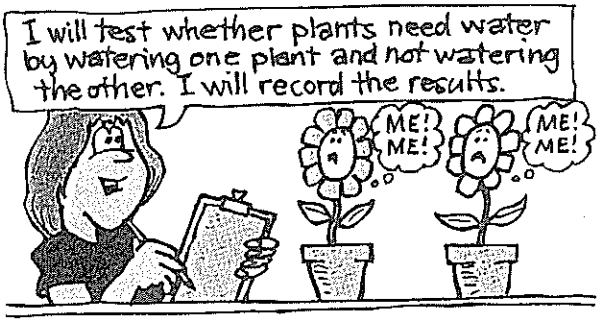
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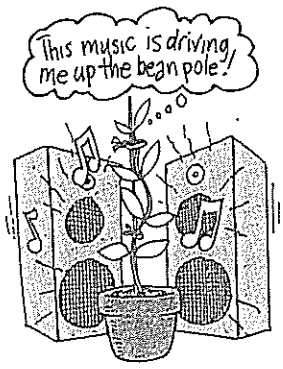


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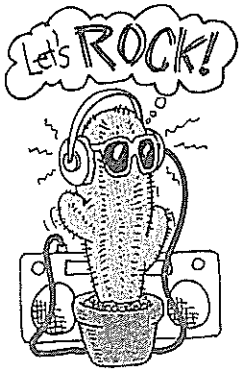


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Exercise 4 Explaining



Anthony plants beans and plays rock music to them for 3 hours a day. The beans grow 15 cm in 2 weeks.



Debbie plants a cactus and plays rock music to it for 10 hours a day. It grows 1 cm in 6 months.



Jessica plants roses and plays Christmas carols to them for 2 hours a day. The roses die within a month.

1 Does this mean that roses hate Christmas carols or that beans like rock music more than a cactus does? ..... Why or why not? .....
.....

2 If you want to find out if plants respond to music (a) What must you keep the same?
.....
(b) What must you change?.....
.....

3 Why is it a good idea to keep at least one plant which does not have music played in its environment? .....
.....

Exercise 5 Hypothesising (or writing a hypothesis)

1 A hypothesis is a generalisation which can be tested. Tick the generalisations below which you could test in an experiment. Discuss, in a group, how you would test each hypothesis.
(a) Heavy people go faster down a waterslide.
(b) Cats are usually more intelligent than dogs.
(c) Women are generaly better drivers than men.

2 Complete the table below.

OBSERVATIONS	David can lift heavier weights than Alison. Carlos can lift heavier weights than Beth. Stefan can lift heavier weights than Mary-Louise. Mr Bloggs can lift heavier weights than Mrs Hogan.
GENERALISATION (HYPOTHESIS)	....
PREDICTION	....

Exercise Matching illustrations and captions

1 The eight steps are illustrated *in the wrong order* below. Write the correct steps from the previous page in the space beneath each illustration. The first step is done for you.

